DSA 495 601 Project Milestone 2

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- Dataset Selection
- Project Scoping
- Read & Explore Your Dataset
- Questions About the Data

For Milestone 2, select your dataset for the project and then complete this markdown. Knit your completed file to a HTML. Download the HTML to your computer and then print or save it to a PDF file with the following naming convention: course_section_semester_assignment_unityID.pdf. For example, DSC495_601_FA25_Milestone2_keporte2.pdf.

Due: Septebmer 18 before class.

Dataset Selection

Question: What dataset did you choose? *(Provide the file name or brief description)*

I chose the dataset: `bank-full.csv`
This dataset contains data from telephone marketing campaigns of a Portuguese Bank. Each row represents a contact with a client. There are 17 variables, one of them being a variable “y”, which is our target. It references whether or not a client has agreed to open a term deposit account, and the variable takes values “Yes” and “No”.

Project Scoping

(a) What is the **binary outcome** you plan to predict?

The binary outcome I’m predicting is `y` — which represents whether or not a client has subscribed to a term deposit. The variable `y` takes character values “Yes” and “No”. For modeling, I am converting this to a binary variable with (“Yes” = 1) and (“No” = 0).

(b) Who is your **target population**?

My target population is all customers in the current marketing campaign. We will use data from all the customers and variables in the dataset, so there is no need for specific filtering.

(c) **When** are you making the prediction?

We will be making predictions throughout the duration of the marketing campaign. All available information on a customer will be utilized to make predictions, and predictions will be made throughout the entire campaign

(d) Why would **predictive analytics** be valuable here?

Predictive analytics is valuable here because it gives the bank the opportunity to spend its marketing resources more effectively. We are identifying in real time, during a marketing campaign, which customers are most likely to end up subscribing. This gives the bank valuable insight about how to follow-up and continue the campaign.

Read & Explore Your Dataset

Load the necessary packages

```
.libPaths(c("/share/dsa495601f25/installed_libraries", .libPaths()))
library(dplyr)
library(skimr)   # For easy summaries
library(janitor) # For formatting object names
```

Read in your data

```
data_file <- paste0(params$data_path,"bank-full.csv")
df <- read.csv(data_file, sep = ";") %>% janitor::clean_names()
```

How many rows and columns?

```
dim(df)  # Returns rows and columns
```

```
## [1] 45211    17
```

Make the outcome binary (if needed)

```
df <- df %>%
  mutate(y_binary = if_else(y == "yes", 1, 0))
```

Check distribution of the binary outcome

```
df %>%
  count(y_binary) %>%
  mutate(percent = n / sum(n) * 100)
```

y_binary <dbl>	n <int>	percent <dbl>
0	39922	88.30152
1	5289	11.69848

2 rows

Get a summary of your data

```
# Glimpse for a compact preview of the data
glimpse(df)
```

```
## Rows: 45,211
## Columns: 18
## $ age       <int> 58, 44, 33, 47, 33, 35, 28, 42, 58, 43, 41, 29, 53, 58, 57, …
## $ job       <chr> "management", "technician", "entrepreneur", "blue-collar", "…
## $ marital   <chr> "married", "single", "married", "married", "single", "marrie…
## $ education <chr> "tertiary", "secondary", "secondary", "unknown", "unknown", …
## $ default   <chr> "no", "no", "no", "no", "no", "no", "no", "yes", "no", "no",…
## $ balance   <int> 2143, 29, 2, 1506, 1, 231, 447, 2, 121, 593, 270, 390, 6, 71…
## $ housing   <chr> "yes", "yes", "yes", "yes", "no", "yes", "yes", "yes", "yes"…
## $ loan      <chr> "no", "no", "yes", "no", "no", "no", "yes", "no", "no", "no"…
## $ contact   <chr> "unknown", "unknown", "unknown", "unknown", "unknown", "unkn…
## $ day       <int> 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, 5, …
## $ month     <chr> "may", "may", "may", "may", "may", "may", "may", "may", "may…
## $ duration  <int> 261, 151, 76, 92, 198, 139, 217, 380, 50, 55, 222, 137, 517,…
## $ campaign  <int> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, …
## $ pdays     <int> -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, -1, …
## $ previous  <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
## $ poutcome  <chr> "unknown", "unknown", "unknown", "unknown", "unknown", "unkn…
## $ y         <chr> "no", "no", "no", "no", "no", "no", "no", "no", "no", "no", …
## $ y_binary  <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, …
```

```
# Skim for a more in depth overview
skim(df)
```

Data summary

Name	df
Number of rows	45211
Number of columns	18
_____	
Column type frequency:	
character	10
numeric	8
_____	
Group variables	None

Variable type: character

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
job	0	1	6	13	0	12	0
marital	0	1	6	8	0	3	0
education	0	1	7	9	0	4	0

skim_variable	n_missing	complete_rate	min	max	empty	n_unique	whitespace
default	0	1	2	3	0	2	0
housing	0	1	2	3	0	2	0
loan	0	1	2	3	0	2	0
contact	0	1	7	9	0	3	0
month	0	1	3	3	0	12	0
poutcome	0	1	5	7	0	4	0
y	0	1	2	3	0	2	0

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
age	0	1	40.94	10.62	18	33	39	48	95	
balance	0	1	1362.27	3044.77	-8019	72	448	1428	102127	
day	0	1	15.81	8.32	1	8	16	21	31	
duration	0	1	258.16	257.53	0	103	180	319	4918	
campaign	0	1	2.76	3.10	1	1	2	3	63	
pdays	0	1	40.20	100.13	-1	-1	-1	-1	871	
previous	0	1	0.58	2.30	0	0	0	0	275	
y_binary	0	1	0.12	0.32	0	0	0	0	1	

Questions About the Data

What questions do you have about the data at this time?

Some questions I have include:

Since we have such a great class imbalance between our binary y values (11.698% = “Yes”), how will our training and validation metrics be affected? Should we use methods like weighting to try and address this issue?

Which categorical variables will be especially useful in prediction? How can we feature engineer these variables to make useful quantifiable variables to use in our model?

How impactful is the temporal aspect of the data? Would predictive information from a campaign in the summer be useful in a winter campaign?